19/03/2021

Code-A

Aakash

Medical | IIT-JEE | Foundations

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MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - 14

Answer Key

1. (2)	37. (2)	73. (2)	109. (1)	145. (4)
2. (2)	38. (3)	74. (3)	110. (3)	146. (2)
3. (1)	39. (2)	75. (2)	111. (2)	147. (4)
4. (2)	40. (4)	76. (2)	112. (4)	148. (3)
5. (4)	41. (1)	77. (3)	113. (2)	149. (1)
6. (1)	42. (3)	78. (2)	114. (1)	150. (2)
7. (2)	43. (4)	79. (1)	115. (4)	151. (2)
8. (3)	44. (3)	80. (4)	116. (3)	152. (4)
9. (3)	45. (1)	81. (4)	117. (3)	153. (2)
10. (2)	46. (2)	82. (2)	118. (2)	154. (1)
11. (2)	47. (3)	83. (4)	119. (3)	155. (2)
12. (4)	48. (3)	84. (4)	120. (4)	156. (3)
13. (4)	49. (3)	85. (1)	121. (3)	157. (4)
14. (2)	50. (2)	86. (4)	122. (1)	158. (4)
15. (2)	51. (3)	87. (2)	123. (2)	159. (2)
16. (2)	52. (1)	88. (4)	124. (1)	160. (1)
17. (4)	53. (3)	89. (1)	125. (3)	161. (1)
18. (4)	54. (3)	90. (4)	126. (4)	162. (3)
19. (4)	55. (2)	91. (2)	127. (2)	163. (4)
20. (2)	56. (1)	92. (1)	128. (2)	164. (4)
21. (3)	57. (4)	93. (3)	129. (3)	165. (1)
22. (1)	58. (2)	94. (3)	130. (2)	166. (3)
23. (1)	59. (2)	95. (1)	131. (1)	167. (2)
24. (2)	60. (4)	96. (3)	132. (3)	168. (4)
25. (1)	61. (1)	97. (2)	133. (3)	169. (3)
26. (1)	62. (2)	98. (2)	134. (4)	170. (4)
27. (2)	63. (3)	99. (1)	135. (1)	171. (2)
28. (2)	64. (1)	100. (4)	136. (2)	172. (2)
29. (3)	65. (1)	101. (3)	137. (4)	173. (3)
30. (4)	66. (2)	102. (3)	138. (2)	174. (2)
31. (3)	67. (2)	103. (1)	139. (3)	175. (4)
32. (2)	68. (3)	104. (3)	140. (3)	176. (2)
33. (4)	69. (3)	105. (1)	141. (4)	177. (1)
34. (4)	70. (3)	106. (4)	142. (1)	178. (3)
35. (1)	71. (3)	107. (2)	143. (4)	179. (2)
36. (3)	72. (1)	108. (3)	144. (1)	180. (3)

62. Answer (2)

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

2	0	0
2×0.6	2×0.4	2×0.4
$=1.2$	0.8	0.8

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]} = \frac{\left(\frac{0.8}{10}\right)\left(\frac{0.8}{10}\right)}{\left(\frac{1.2}{10}\right)} = \frac{0.8\times0.8}{1.2\times10}$$

$$= 0.0533 = \frac{4}{75}$$

63. Answer (3)

$$[H^+]_{Final} = \frac{0.1}{1000} = 10^{-4}$$

$pH = -\log[H^+] = 4$

64. Answer (1)

Mixture of weak acid and its salt with strong base form an acidic buffer.

65. Answer (1)

Ag_2CO_3 is most soluble due to having maximum solubility among given

$Ag_2CO_3 \rightleftharpoons 2Ag^+ + CO_3^{2-}$

$K_{sp} = (2s)^2(s) = 4s^3 = 8 \times 10^{-12}$

$s = (2)^{1/3} \times 10^{-4} = 1.26 \times 10^{-4}$

66. Answer (2)

Reducing agent, H_2O_2 converts I_2 into I^-.

67. Answer (2)

$E^\circ_{Cell} = E^\circ_{Cathode} - E^\circ_{Anode}$

$= 1.4 - (-0.13) = 1.4 + 0.13 = 1.53$ V

68. Answer (3)

- Volume strength = 11.2 × Molarity (H_2O_2)

Molarity (H_2O_2) = $\frac{16.8}{11.2} = 1.5$ M

- % strength = $\frac{1.5\times34}{10} = 5.1$

69. Answer (3)

Potassium on reaction with excess air gives superoxide.

70. Answer (3)

LiCl hydrated as $LiCl.2H_2O$ while $MgCl_2$ hydrated as $MgCl_2.8H_2O$

71. Answer (3)

Compound containing at least two α-Hydrogen atoms w.r.t. carbonyl group can give aldol condensation product.

72. Answer (1)

Primary amine (only aliphatic) can be prepared by Gabriel phthalimide process.

73. Answer (2)

$\Delta_f H^\circ$ of graphite is taken as zero. Therefore it is the most stable allotrope of carbon.

74. Answer (3)

$3B_2H_6 + 6NH_3 \longrightarrow 3[BH_2(NH_3)_2]^+[BH_4]^-$

$\downarrow$ Heat

$2B_3N_3H_6 + 12H_2$

75. Answer (2)

$\Delta T_f = iK_f m$

$\{i = 1 + (4-1)\times0.95 = 1 + 3\times0.95 = 3.85\}$

$\Delta T_f = 3.85 \times 1.86 \times 0.2 = 1.43$

Freezing point of solution

$= 273 - 1.43 = 271.57$

76. Answer (2)

$$\frac{d_{bcc}}{d_{sc}} = \frac{\left(\frac{ZM}{N_A \times a^3}\right)_{bcc}}{\left(\frac{ZM}{N_A a^3}\right)_{sc}} = \frac{\frac{4}{(2\sqrt{2})^3}}{\left(\frac{1}{2}\right)^3} = \frac{\frac{4}{16\sqrt{2}}}{\left(\frac{1}{8}\right)} = \sqrt{2} : 1$$

77. Answer (3)

$2H^+(aq) + 2e^- \longrightarrow H_2(g)$

$$E = \frac{-0.0591}{2}\log\left(\frac{pH_2}{[H^+]^2}\right)$$

$= +0.0591 \log[H^+] = -0.0591\ pH = 0$

78. Answer (2)

$CuCl_2(aq) \longrightarrow Cu^{2+}(aq) + 2Cl^-(aq)$

Cathode : $Cu^{2+} + 2e^- \longrightarrow Cu(s)$

Anode : $2Cl^- \longrightarrow Cl_2 + 2e^-$

79. Answer (1)

(I) $k = \frac{2.303}{30}\log\left(\frac{1.2}{0.3}\right) = \frac{2.303}{30}\log 4$

(II) $k = \frac{2.303}{t}\log\left(\frac{1.8}{0.45}\right) = \frac{2.303}{t}\log 4$

Hence, t = 30 minutes.

80. Answer (4)

Methylene blue sol being positive is easily coagulated by $[Fe(CN)_6]^{3-}$ ion.

81. Answer (4)

Zirconium is refined by van Arkel method (vapour phase refining)

82. Answer (2)

In contact process V_2O_5, is used as catalyst to convert SO_2 into SO_3.

83. Answer (4)

$XeF_4 + O_2F_2 \longrightarrow XeF_6 + O_2$

84. Answer (4)

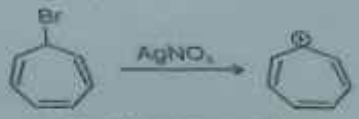
Yb^{3+} : [Xe] $4f^{13}$

85. Answer (1)

Co^{3+} : [Ar] $3d^6$ — e_g, t_{2g}

$t_{2g}^6\ e_g^0$

86. Answer (4)



Aromatic compound

87. Answer (2)

Ring of highest electron density gives fastest rat of electrophilic aromatic substitution.

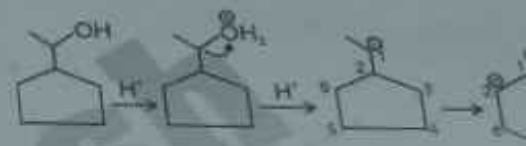
88. Answer (4)

Molecule having no centre of symmetry and plane of symmetry, can be optically active.

89. Answer (1)

Conjugation with carbonyl group stabilises transition state in S_N2 mechanism. Therefor more reactive towards S_N2.

90. Answer (4)



BOTANY

91. Answer (2)

Hydrilla and *Zostera* – Pollination is hydrophily

92. Answer (1)

Pea, groundnut, beans

93. Answer (3)

The term clone is applicable to the offspring formed by vegetative reproduction.

94. Answer (3)

There will be no segregation in hybrid progeny.

95. Answer (1)

AUG codes for methionine and also act as initiator codon.

96. Answer (3)

4 phenotypes (9 : 3 : 3 : 1)

9 genotypes (1 : 2 : 2 : 4 : 1 : 2 : 1 :

97. Answer (2)

Marchantia is dioecious specie separate female and male thallus.

98. Answer (2)

Wildlife sanctuary and sacred g conservation. Zoo is – ex-s Endemism – Species found to t found anywhere else.

99. Answer (1)

Gibberellic acid and nitrat germination.

100. Answer (4)

Viroids have only RNA protein are absent.

33. Answer (4)

$B = 5t$

$A = \pi R^2$, $\phi = BA\cos\theta$

$\phi = BA$ [As angle between B and A is either 0° or 180° so magnitude of flux will remain same]

$= (5t)(\pi R^2)$

$e = \left|\frac{d\phi}{dt}\right| = \pi R^2 \times (10t)$

$i = \frac{e}{r} = \left(\frac{10\pi R^2}{r}\right)t$

34. Answer (4)

We Know, potential difference $= \frac{B\omega L^2}{2}$

Now, $V_D - V_A = \frac{B_0\omega\left(\frac{L}{4}\right)^2}{2} = \frac{B_0\omega L^2}{32}$ ...(i)

$V_D - V_B = \frac{B_0\omega\left(\frac{L}{2}+\frac{L}{4}\right)^2}{2} = \frac{B_0\omega\left(\frac{2L+L}{4}\right)^2}{2}$ ...(ii)

$= \frac{B_0\omega\left(\frac{9L^2}{16}\right)}{2} = \frac{9B_0\omega L^2}{32}$

Equation (ii) − (i)

$V_D - V_B + V_D - V_A = \frac{8}{32}B_0\omega L^2 - \frac{B_0\omega L^2}{32}$

$V_A - V_B = \frac{8}{32}B_0\omega L^2 - \frac{B_0\omega L^2}{4}$

35. Answer (1)

$\omega^2 = 16$

$\omega = 4 = 2\pi f$

$f = \frac{2}{\pi}$

36. Answer (1)

$\frac{1}{2}LI^2 = \frac{Q^2}{2C}$

$\Rightarrow \frac{1}{2}L\left(\frac{E}{R}\right)^2 = \frac{Q^2}{2C}$

$\Rightarrow Q = \sqrt{LC}\left(\frac{E}{R}\right)$

37. Answer (2)

$V = 10\sin(50t + 10)$

$X_C = \frac{1}{\omega C} = \frac{10^6}{50\times 3}$

Now

$I = I_0\sin(50t + 10° + 90°)$

$= 15 \times 10^{-6}\sin(50t + 100°)$

38. Answer (3)

$Z = \sqrt{R^2 + X_L^2} = \sqrt{9+16} = \sqrt{25} = 5\,\Omega$

39. Answer (2)

Average energy density of electric field = average energy density of magnetic field

$u_E = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0\left(\frac{E_0}{\sqrt{2}}\right)^2 = \frac{1}{4}\varepsilon_0 E_0^2$

$= \frac{1}{4}\times 8.85\times 10^{-12}(1)^2 = 2.2\times 10^{-12}\ J/m^3$

40. Answer (4)

$V = \frac{C}{1.5} = 2\times 10^8\ m/s$

$V = f\lambda$

$\lambda = \frac{2\times 10^8\ m/s}{5\times 10^{14}\ Hz}$

$\lambda = 4\times 10^{-7}$

$N = \frac{10^{-3}}{4\times 10^{-7}} = 2500$

41. Answer (1)

Dispersive power is defined as angular Dispersion per unit mean deviation.

42. Answer (3)

The intensity will be minimum if path difference is odd integral multiple of half of wavelength i.e. 200 nm and will be maximum if path difference is integral multiple of wavelength i.e. 400 nm.

Thus the resultant intensity will be between minimum and maximum possible values.

43. Answer (4)

de-Broglie wavelength $= \frac{h}{p}$

Thus the ratio of wavelengths remains same and equal to 2: 1

20. Answer (2)

Irreversible engine have less efficiency due to the dissipative effects while a reversible engine consisting of reversible processes which are quasi-static and non-dissipative.

21. Answer (2)

Total energy remains constant in an undamped SHM and is equal to 0.5 kA²

22. Answer (1)

$$y = 2\sin\left(\frac{\pi}{12}t + \frac{\pi}{6}\right)$$

$$V = \frac{dy}{dt} = 2\cos\left(\frac{\pi}{12}t + \frac{\pi}{6}\right)\times\frac{\pi}{12}$$

$$= \frac{\pi}{6}\cos\left(\frac{\pi}{12}t + \frac{\pi}{6}\right)$$

at $t = 2$ sec.

$$v = \frac{\pi}{6}\cos\left(\frac{\pi}{3}\right) = \frac{\pi}{12}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}\times1\times\left(\frac{\pi}{12}\right)^2 = \frac{\pi^2}{288}J$$

23. Answer (1)

In damped oscillation of the body, amplitude of oscillation and energy decreases with time.

24. Answer (2)

$$E = E_0e^{-\alpha t/m}$$

$$\frac{E_0}{2} = E_0e^{-\alpha t/m}$$

$$e^{\alpha t/m} = \frac{1}{2}$$

$$E = E_0e^{-\alpha t/m}$$

$$E = E_0\left(e^{-\alpha t/m}\right)^3$$

$$E = E_0\left(\frac{1}{2}\right)^3 = \frac{E_0}{8}$$

25. Answer (1)

$$S = \frac{1}{2}at^2$$

$$a = F/m = qE/M$$

$$a = 1.6\times10^{-19}\times2.21\times10^{-4}/2\times8.85\times10^{-12}\times1.67\times10^{-27}$$

$$4 = 3.536\times10^{38}t^2/9.559\times10^{-39}$$

$$a = 0.12\times10^{77}$$

$$S = 0.1 = (1/2)(0.12\times10^{77})t^2$$

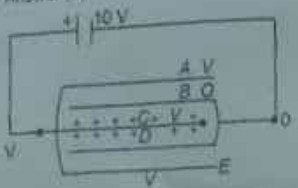
$$t^2 = 1.6\times10^{-77-77} = 16\times10^{-155}$$

$$t = 4\times10^{-78}\text{ s}$$

26. Answer (1)

As $\vec{x}\perp\vec{E} \Rightarrow \Delta V = 0$

27. Answer (2)



Charge on plate C(both side) = CV + CV = 2CV

$Q = 2\times2\times10^{-6}\times10 = +40\ \mu C$

28. Answer (2)

$I = dq/dt$

$I = -4 + 2\sin2t$

$= -8\sin2t$

Maximum value of current = 8 A

[$\because (\sin 2t)_{max} = 1$]

29. Answer (3)

Current, $i = dQ/dt$

$i = 2at - 3bt^2$

Current is maximum or minimum when $di/dt = 0$

$2a - 6bt = 0$

$t = 2a/6b = a/3b$

30. Answer (4)

When the current carrying wire is along magnetic field, then the force on the wire is zero.

31. Answer (3)

As magnetic force is a result of cross product of velocity and magnetic field, it is perpendicular to both. When the angle between velocity and magnetic field is zero of 180° then the magnetic force is zero.

32. Answer (2)

As $F = qvB$

Thus $F \propto v$. This means that the graph will be straight line, inclined at an angle to speed axis.

44. Answer (3)

There will be a gain in overall binding energy when we move from a heavy nuclei region to the middle region of the plot. Hence third statement is incorrect.

CHEMISTRY

46. Answer (2)

$$\text{Molality} = \frac{x_{solute}\times1000}{x_{solvent}\times\text{Molar mass of solvent}}$$

$$= \frac{0.25\times1000}{0.75\times18} = 18.52\text{ m}$$

47. Answer (3)

Aqueous solution of Borax is used as antiseptic.

48. Answer (3)

$$\text{Wave number } (\bar{\nu}) = \frac{1}{\lambda} = \frac{1}{5000\times10^{-8}}\text{cm}^{-1}$$

$$= \frac{10^8}{5}\text{cm}^{-1} = 2\times10^4\text{ cm}^{-1}$$

49. Answer (3)

Mg (12) $1s^2\ 2s^2\ 2p^6\ \boxed{3s^2}$

$n = 3$
$l = 0$
$m = 0$
$s = +\frac{1}{2}$ or $-\frac{1}{2}$

50. Answer (2)

Negative electron gain enthalpies of halogens are highest among p-Block elements.

51. Answer (3)

Percentage of chlorine

$$= \frac{35.5\times0.14\times100}{143.5\times0.35} = 9.89\%$$

52. Answer (1)

Glycine does not has chiral centre, so achiral in nature.

53. Answer (3)

Excess Pb (> 50 ppb) in drinking water damages kidney, brain and liver.

101. Answer (3)

Cells of the tapetum secrete sporopollenin.

102. Answer (3)

RuBisCO activity occurs in chloroplasts of bundle sheath cells in C_4-plants.

103. Answer (1)

High dose of UV-B causes show blindness.

104. Answer (3)

The biomass of a trophic level depends upon reproductive potential and longevity of its members.

105. Answer (1)

Alfred Hershey and Martha Chase.

106. Answer (4)

F_1 resembles either of the two parents in dominance.

F_1 is in between the two parents in incomplete dominance.

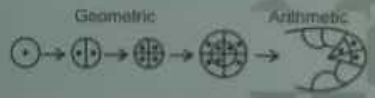
107. Answer (2)

Human beings and *Drosophila* = XY type, female heterogamety is in birds.

108. Answer (3)

Sexual or perfect stage is absent in deuteromycetes.

109. Answer (1)



Growth of embryo development.

110. Answer (3)



111. Answer (2)

$\psi_w = \psi_S + \psi_P$; water movement from higher ψ_w to lower ψ_w.

112. Answer (4)

Sparrow as predator of seeds (Plants).

113. Answer (2)

Middle lamella is made of calcium and magnesium pectate.

114. Answer (1)

The synthesis of lipids and steroidal hormones occur in smooth endoplasmic reticulum.

115. Answer (4)

The condensation of chromosomes occurs throughout the leptotene. Chromosomes elongate even in telophase I.

116. Answer (3)

These thread like structures are root hairs.

117. Answer (3)

Insectivorous plants, perform photosynthesis as they have green leaves; only some leaves are modified to trap insects to compensate the deficiency of nitrogen.

118. Answer (2)

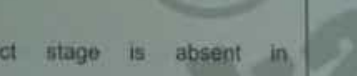
Mesophyll is not differentiated in monocot leaf. Parallel venation is characteristics of monocot.

119. Answer (3)

The primary xylem remains more or less intact, in or around the centre.

120. Answer (4)

Fabaceae's floral formula is



121. Answer (3)

Micropyle of seed is the entry gate of water. Micropyle of ovule/megasporangium is the entry gate of pollen tube.

122. Answer (1)

All options except (1) show actinomorphic flowers.

123. Answer (2)

Meristematic tissues are completely consumed in case of axillary bud and leaf primordium.

124. Answer (1)

Pyruvate	$\xrightarrow{TPP, Mg^{+2}}$	Acetaldehyde + CO_2
Acetaldehyde	$\xrightarrow{NADH}$	Ethyl alcohol + NAD^+

125. Answer (3)

Statins have been commercialised as blood cholesterol lowering agents.

126. Answer (4)

Semi-dwarf wheat was developed by Nobel laureate Norman E. Borlaug.

127. Answer (2)

The two strands bind together because of their hydrogen bonding affinity for each other.

128. Answer (2)

Removal of introns and joining of exons is called splicing.

129. Answer (3)

Lac-operon – Negative regulation by repressor.

UTR enhances translation.

130. Answer (2)

Sigma factor – is a small protein (sub unit) of bacterial RNA polymerase requires for selection of binding sites (Promoter).

131. Answer (1)

Taxon canidae is family, rest are order.

132. Answer (3)

Ectocarpus, *Polysiphonia* and kelps exhibit haplo-diplontic life cycle.

133. Answer (3)

Pinnately compound leaves are found in *Cycas*.

134. Answer (4)

Dryopteris, *Pteris* and *Adiantum* belong to class Pteropsida.

135. Answer (1)

Formation of interzonal fibres occurs. Then separated daughter/sister chromatids start moving towards the opposite poles.

ZOOLOGY

36. Answer (2)

Smooth muscle fibres are uninucleated, non-striated and unbranched.

7. Answer (4)

Electrical synapse, Single unit smooth muscle and cardiac muscle possesses gap junctions.

3. Answer (2)

Arachidic acid : 20C : 0

Stearic acid : 18C : 0

. Answer (3)

Glutamic acid is an acidic amino acid

Answer (3)

Oxaloacetate and malonate resembles succinate in structure and compete for binding with enzyme succinic dehydrogenase.

Answer (4)

Paneth cells are present in intestinal gland and ecrete lysozyme.

nswer (1)

ross calorific value of proteins, fats and rbohydrate is 5.65, 9.45 and 4.1 Kcal/g while ysiological value is 4, 9 and 4 Kcal/g spectively.

swer (4)

C = RV + ERV + TV + IRV

144. Answer (1)

Rise in BPG levels, H^+ concentration (fall in pH) and temperature shifts oxygen dissociation curve towards the right side.

145. Answer (4)

In ventricular systole, blood flows from left and right ventricles to their respective blood vessels.

146. Answer (2)

Cardiac output = Stroke volume × Heart rate

= 70 ml × 70/min = 4900 mL / min.

147. Answer (4)

'B' blood group – B antigen on RBCs, anti-A antibodies in plasma.

'O' blood group – No antigen on RBC's surface

148. Answer (3)

Cellular modification of Distal Convoluted Tubule (DCT) and afferent arteriole constitutes Juxtaglomerular Apparatus (JGA).

149. Answer (1)

Angiotensinogen is converted to Angiotension I by renin released by JG cells. Angiotension II, being a powerful vasoconstrictor increases GFR. It also activates adrenal gland to release aldosterone.

150. Answer (2)

Floating ribs are 2 pairs in number.

151. Answer (2)

Knee joint is an example of hinge joint. Saddle joint is present between carpal and metacarpal of human thumb.

152. Answer (4)

$Na^+ - K^+$ ATPase maintains resting membrane potential, by transporting $3Na^+$ outwards and $2K^+$ into the cell.

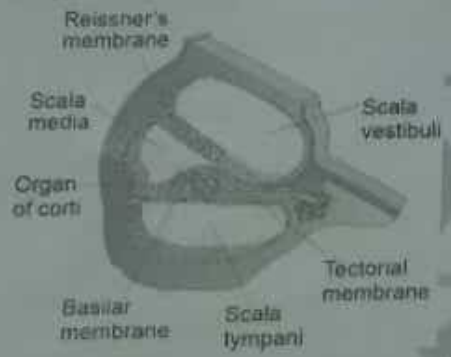
153. Answer (2)

Cerebellum is a part of hindbrain.

154. Answer (1)

Fovea is a thinned out portion of the retina where only the cones are densely packed.

155. Answer (2)



Scala media is filled with endolymph while scala vestibule and tympani with perilymph.

156. Answer (3)

Life span of crow, parrot, crocodile and tortoise is 15 years, 140 years, 60 years and 100–150 years respectively.

157. Answer (4)

Zygote formation is universal in all sexually reproducing organisms.

158. Answer (4)

Earthworm do not exhibit parthenogenesis.

159. Answer (2)

Chromosome number in gamete of housefly is 6.

160. Answer (1)

Transfer of sperms into the female genital tract is called insemination.

161. Answer (1)

Each testis has 250 compartments

∴ Testes = 500 compartments

1-3 seminiferous tubules / compartment

∴ 500–1500 seminiferous tubules

162. Answer (3)

Oviduct is fallopian tube. Isthmus connects oviduct to uterine fundus.

163. Answer (4)

After spermiogenesis, sperms are finally released from seminiferous tubules by the process called spermiation.

164. Answer (4)

Progesterone attain their peak in middle of luteal phase.

165. Answer (1)

Saheli, a new oral contraceptive for the females, was developed at CDRI.

Nirodh is male condom.

166. Answer (3)

Increase in number of people in reproducible age increases population growth rate.

167. Answer (2)

Vault is a barrier method while Lippes loop is an inert IUD.

168. Answer (4)

GIFT : Gamete Intrafallopian Transfer
AI : Artificial Insemination
IUT : Intrauterine Transfer
ZIFT : Zygote Intrafallopian Transfer

169. Answer (3)

Closed flask contains CH_4, H_2, NH_3, water vapour at 800°C.

170. Answer (4)

Homology is based on divergent evolution. Options (1), (2) and (3) are examples of analogy which is based on convergent evolution.

171. Answer (2)

Lemur and spotted cuscus is an example of convergent evolution.

172. Answer (2)

Coelacanth, a fish caught in South Africa.

7. Answer (3)

When the block compresses the spring, let x m be the amount of compression (i.e. $x = 4$ m). Let v = velocity of the block when it collides with the spring.

Loss in K.E. of the block = (gain in elastic potential energy of the spring) + (work done against friction)

$$\Rightarrow \frac{1}{2}mv^2 - 0 = \frac{1}{2}kx^2 + \mu mgx$$

$$\Rightarrow v = \sqrt{32 + 8} = 7.16 \text{ m/s}$$

8. Answer (3)

Angular impulse = Torque × Time = 6 × 3

= 18 kg m²/s

$I\omega = 18$

$6\omega = 18$

$\omega = 3$ rad/s

15. Answer (4)

$$T = \frac{2m_1m_2}{m_1 + m_2}g = \frac{2 \times 1 \times 2}{1+2} \times 10 = \frac{40}{3} \text{ N}$$

If r is the minimum radius, then:

$$\text{Breaking stress} = \frac{\frac{40}{3}}{\pi r^2} \text{ or } \frac{40}{3\pi r^2} \times 10^8 = \frac{40}{3\pi r^2}$$

$$\text{or } r^2 = \frac{1}{10^8} \text{ or } r = \frac{1}{10^4} \text{ m}$$

$$\text{or } r = \frac{1}{10^4} \times 10^3 \text{ mm} = 1 \text{ mm}$$

16. Answer (2)

When a force F is applied at the cross-section area A of a wire, then

$$\text{Stress} = \frac{F}{A} = 4.0 \times 10^7 \text{ N/m}^2$$

17. Answer (3)

$a = 0.12 \times 10^{11}$

$S = 0.1 = (1/2)(0.12 \times 10^{11})t^2$

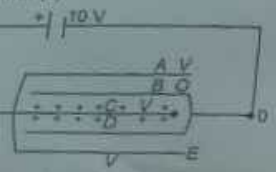
$t = 1.6 \times 10^{-11} = 16 \times 10^{-12}$

$= 4 \times 10^{-6}$ s

swer (1)

$x \perp \vec{E} \Rightarrow \Delta V = 0$

swer (2)



s on plate C(both side) = $CV + CV = 2CV$

$2 \times 10^{-6} \times 10 = +40\ \mu C$

(2)

$2 \sin 2t$

2^t

value of current = 8 A

$[\because (\sin 2t)_{max} = 1]$

dQ/dt

t^2

maximum or minimum when

$\sqrt{3}b$

rent carrying wire is along magnetic force on the wire is zero.

rce is a result of cross product of gnetic field, it is perpendicular to e angle between velocity and s zero of 180° then the magnetic

means that the graph will be ed at an angle to speed axis.

44. Answer (3)

There will be a gain in overall binding energy when we move from a heavy nuclei region to the middle region of the plot. Hence third statement is incorrect.

45. Answer (1)

Only in common emitter configuration both gains are higher than unity so common emitter configuration is preferred for making amplifier.

CHEMISTRY

46. Answer (2)

$$\text{Molality} = \frac{x_{solute} \times 1000}{x_{solvent} \times \text{Molar mass of solvent}}$$

$$= \frac{0.25 \times 1000}{0.75 \times 18} = 18.52\ m$$

47. Answer (3)

Aqueous solution of Borax is used as antiseptic.

48. Answer (3)

$$\text{Wave number } (\bar{\nu}) = \frac{1}{\lambda} = \frac{1}{5000 \times 10^{-8}} cm^{-1}$$

$$= \frac{10^8}{5} cm^{-1} = 2 \times 10^8 cm^{-1}$$

49. Answer (3)

Mg (12) $1s^2\ 2s^2\ 2p^6\ \boxed{3s^1}$

$n = 3$
$l = 0$
$m = 0$
$s = +\frac{1}{2}$ or $-\frac{1}{2}$

50. Answer (2)

Negative electron gain enthalpies of halogens are highest among *p*-Block elements.

51. Answer (3)

Percentage of chlorine

$$= \frac{35.5 \times 0.14 \times 100}{143.5 \times 0.35} = 9.89\%$$

52. Answer (1)

Glycine does not has chiral centre, so achiral in nature.

53. Answer (3)

Excess Pb (> 50 ppb) in drinking water damages kidney, brain and liver.

54. Answer (3)

Both $N_2(13e^-)$ and $N_2^-(15e^-)$ are having one unpaired electron, hence having same magnetic property.

55. Answer (2)

XeF_4 has square planar shape, in which all atoms are in same plane.

56. Answer (1)

For strongest H-Bond, covalently attached hydrogen atom should be attached with most electronegative atom.

57. Answer (4)

With increase in effective nuclear charge ionization energy increases.

58. Answer (2)

$$d = \frac{PM}{RT}$$

$$d = \frac{1\ atm \times 64\ g\ mol^{-1}}{0.0821\ L\ atm\ K^{-1}\ mol^{-1} \times 400\ K}$$

$$= \frac{64}{0.0821 \times 400} = 1.95\ g/L$$

59. Answer (2)

ΔH ionization

$= -7.7 - (-13.7) = 13.7 - 7.7 = 6$ kcal

60. Answer (4)

- For isothermal process, $\Delta U = 0$
- From 1st law of thermodynamics

$\Delta U = Q + W$

$Q = -W$

$\because W < 0$ so $Q > 0$

61. Answer (1)

Specific heat capacity is an intensive property.

173. Answer (3)

Ascariasis, filariasis – Helminthic infection

Common cold – Viral disease

Amoebic dysentery – Protozoan disease

174. Answer (2)

Culex mosquito transmits filariasis caused by *Wuchereria*.

175. Answer (4)

Colostrum provides natural passive immunity.

176. Answer (2)

Cocaine is commonly called coke.

Morphine / smack is an opioid.

177. Answer (1)

Hisardale is an example of cross breed.

178. Answer (3)

DNA fragments are negatively charged and move towards anode. Smallest fragment is closest to anode while largest is closest to loading well.

179. Answer (2)

Bam HI site at tetracycline resistant gene, and *Pvu* I site in ampicillin resistance gene become inactivated. So, transformed recombinant will not grow on antibiotic containing medium.

180. Answer (3)

Dolly was a cloned sheep.

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